Lecture 6.4: Experiments, Qualitative Evidence, and Inference

AREC 570 / ECON 530 · Unit 6

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Before class

Read

- Gibson, “Are You Estimating the Right Thing? An Editor Reflects” (Gibson 2019)
- Kim, “Tackling False Positives in Business Research” (Kim 2019)

Additional readings on experiments and qualitative research

- Falk and Heckman on laboratory experiments (Falk and Heckman 2009)
- Starr on qualitative and mixed methods (Starr 2014)

Readings and course handouts are available on Canvas

Methods covered in Unit 6

- **6.1** Used reduced-form comparisons and developed an estimating equation and identifying assumption from a research question
- **6.2** Estimated structural parameters from a model of behavior and used the model to examine counterfactuals
- **6.3** Used machine learning for prediction and within other empirical analyses

6.4 covers experimental design, qualitative and mixed methods, measurement, statistical power, multiple testing, and confidence intervals

Quantitative and qualitative evidence

Approach	Evidence	Research use
Quantitative	Numerical measures from surveys, records, or experiments	Estimate patterns, effects, parameters, or predictions
Qualitative	Interviews, field observations, or documents	Document decisions and the processes that produce them
Mixed methods	Numerical and qualitative evidence	Examine a question using measured outcomes and accounts of decisions

Select a method to match the research task

Research task	Evidence or method
Estimate a policy effect	Random assignment or an observational comparison with a stated identifying assumption
Estimate preferences	Choice data and a model of consumer or producer behavior
Predict a future outcome	A prediction rule evaluated with held-out observations
Document a decision process	Interviews, fieldwork, or case studies
Test a response to an incentive	A field or lab experiment

Qualitative methods

Interviews

Research use: Interviews can document whether farms received and understood the report, how they used it, and which irrigation changes were feasible

Study procedures

- Recruit across farm size, crop, water rights, and location
- Document the interview guide and follow-up questions
- Describe recording, transcription, and coding
- Examine accounts that challenge the proposed explanation

Reporting: State the recruitment criteria and evidence for each interpretation

Source: Starr (2014), Sections 2 and 3

Field experiment (1/2)

Participating irrigation districts want to know whether offering farms a monthly water-use report reduces irrigation water use

Design element	Hypothetical study
Eligible farms	Farms on the districts' customer lists
Assignment	Random offer of the report at the farm level
Outcome	Total monthly water use in acre-feet, collected in a follow-up survey

Field experiment (2/2)

1. Assign eligible farms to the offer or comparison group
2. Survey both groups about subsequent monthly water use
3. Compare group means to estimate the effect of **being offered** the report
4. Document report delivery, survey response, and departures from assignment

The estimate applies to the offer among farms included in the experiment

Extrapolation to other districts or a mandatory reporting program requires evidence about those settings (i.e., generalizability)

Lab experiments

Falk and Heckman's gift-exchange experiment examines whether workers respond to higher wages with higher effort

1. Participants act as firms or workers
2. A firm offers a wage that the worker can accept
3. A worker who accepts sees the wage and chooses an effort level that is costly to provide
4. The experiment specifies the available choices, minimum effort, and payoff rules

Researchers observe wages and effort under these rules and examine whether effort rises with the wage offered

Source: Falk and Heckman (2009), discussion of the gift-exchange experiment

Mixed methods

Mixed methods designs

Two qualitative methods combined in various sequences

Sequence in the water-use study	Contribution
Interviews before the survey	Identify how farms track water use and refine the survey measure
Interviews after the survey	Investigate why some farms used the report and others did not
Concurrent interviews	Compare accounts of irrigation changes with reported water use

The interviews examine how farms responded to the report. The survey comparison estimates the effect of offering it

Source: Starr (2014), Section 2

Research method concepts

Four common statistical terms

Population

- Definition: Units of interest
- Example: Farms on district customer lists

Measure

- Definition: Recorded indicator of a concept
- Example: Survey-reported water use

Comparison

- Definition: Groups whose outcomes are compared
- Example: Farms offered a report and farms not offered one

Limits

- Definition: Conditions narrowing the conclusion
- Example: Missing survey responses or unlisted farms

Population represented in the data

- **Included:** Farms served by participating districts appear on the customer lists
- **Possibly excluded:** Farms relying entirely on private wells or other suppliers may be absent
- **Population claim:** An estimate for farms on the lists requires additional evidence before applying it to all farms in the region

Gibson's distinction between **registered** and **resident** households illustrates how a data source can omit members of the population named in the research question

Source: Gibson (2019), pages 331 to 332

Measurement and interpretation

The outcome of interest is **total irrigation water use**. A survey question about **district deliveries** records only one source of water. Withdrawals from private wells would be omitted

If farms substitute groundwater for district water after receiving the report, district deliveries could decline while total water use remains unchanged

Gibson's food example makes the same point: one week's purchases omit food consumed from household stocks

Source: Gibson (2019), discussion of food purchases and consumption

Estimate from the assigned groups

The analysis subtracts mean water use in the comparison group from mean water use in the group offered the report

$$\hat{\tau} = \bar{Y}_{\text{offer}} - \bar{Y}_{\text{comparison}}$$

For illustration, suppose this difference is **−2 acre-feet per farm-month**. The negative sign indicates lower reported use among farms offered the report

With comparable follow-up in both assigned groups, the difference estimates the effect of the offer

Statistical inference and sample planning

Random variation and survey response

Farms were randomly assigned to the report or comparison group. The follow-up survey records water use only for farms that respond. Before interpreting the difference between groups, ask

1. **Random variation:** Could another assignment produce a different group difference? A confidence interval or test quantifies this uncertainty
2. **Selective response:** Do respondents differ from farms that did not answer? Compare response rates and available farm characteristics by assigned group

From an estimated difference to a conclusion

In this hypothetical study, responding farms offered the report use **2 acre-feet less** than responding comparison farms. What would justify a claim that the report changes water use?

- **Test level:** Limit the chance of declaring an effect when there is none (Type I error)
- **Power:** Collect enough outcomes to detect a reduction large enough to matter (reduce Type II error)
- **Confidence interval:** Show which effect sizes are consistent with the observed difference
- **Response check:** Assess whether missing surveys change the group comparison

The statistical goal is to limit false positives while making meaningful effects detectable. Report the estimate and its uncertainty alongside the response process

False positives, false negatives, and power

Test the null hypothesis of **no effect of the offer** against a specified alternative, such as a reduction of **2 acre-feet per farm-month**

Decision	**No effect is present**	**A 2-unit reduction is present**
Reject no effect	False positive (Type I error)	Detect the reduction
Do not reject	Correct non-rejection	False negative (Type II error)

At a 5% test level, the Type I error rate is 5% under the null.

If the true effect is a 2-unit reduction, a design with **80% power** would reject the no-effect hypothesis in about 80 of 100 repetitions

Source: Kim (2019), Section 2

Sample size depends on the research objective

For a survey that **describes average water use**, choose a sample size based on the desired precision of the estimated mean

For the **report experiment**, choose a sample size that can detect a specified difference between assigned groups. That calculation uses the effect size, variation in water use, test level, and power target

Sources: Lecture 5.1; Kim (2019), Section 3.5

Choosing an effect and power target

Suppose a reduction of **2 acre-feet per farm-month** would justify the cost of the report program. The study plans to detect an effect of that size

An **80% power target** is a common planning convention. If the true effect is a 2-unit reduction, the planned study would detect it in about four out of five repetitions under the assumed design. A 90% target would require more completed observations

The target is chosen before collecting outcomes and depends on the question and feasible sample size

Source: J-PAL, Power calculations

What reviewers examine in a power calculation

There is no single power cutoff for every study. Reviewers examine the basis for the calculation

1. Why the specified effect size matters for the research or policy question
2. How the assumed variation in water use was estimated
3. Whether the calculation reflects assignment, clustering, and expected survey response
4. Whether the eligible farm lists can produce the required completed sample

After data collection, report the realized sample and the precision of the estimated effect

Sources: J-PAL, Power calculations; course empirical strategy checklist

Inputs to the power calculation

The calculation begins with the effect the study needs to detect: **2 acre-feet less water use per farm-month**

- **Outcome variation:** Assume a standard deviation of 8 acre-feet in each group, based on pilot or prior data
- **Test level:** Use a two-sided 5% test
- **Power target:** Plan for an 80% chance of detecting the 2-unit reduction if it is present
- **Design:** Compare equal-sized groups of independently assigned farms

Assignment by district or correlated outcomes within districts would require a calculation that accounts for clustering

Source: Kim (2019), Sections 2 and 3.5

Completed surveys needed for 80% power

With a 2-unit effect, standard deviation of 8, and a two-sided 5% test, an approximate two-group calculation is

$$n \approx \frac{2(8)^2(1.96 + 0.84)^2}{(2)^2} \approx 251$$

Base R's two-sample t calculation gives **252.13**, so the study plans for **253 completed surveys per group**

Illustrative calculation: `power.t.test(delta = 2, sd = 8, sig.level = .05, power = .80, type = "two.sample")`

Planning the survey contacts

The power calculation gives the number of **observed outcomes** needed: 253 farms per group. Because the outcome comes from a follow-up survey, the team must invite more farms than that

Suppose a pilot with similar farms suggests a **40% response rate**

$$\text{Invitations per group} = \left\lceil \frac{253}{0.40} \right\rceil = 633$$

The team would invite **633 farms in each group** (1,266 in total). This calculation checks whether the customer lists and fieldwork budget can supply the planned number of completed outcomes

The 40% response rate is an illustrative pilot assumption, not a general benchmark

If survey response is 8%

At the planned **40% response rate**, 633 invitations yield about **253 completed surveys per group** and 80% power under the stated assumptions

At a realized **8% response rate**, the same invitations yield about **51 completed surveys per group**. Power to detect the 2-unit effect falls to about **24%**

To obtain 253 completions at an 8% rate, the team would need about **3,163 invitations per group**. The customer lists may contain fewer eligible farms

Illustrative calculation: R `power.t.test(n = 51, delta = 2, sd = 8, sig.level = .05, type = "two.sample")`

Nonresponse and interpretation

There is no universal expected response rate. Use a pilot or comparable survey to plan invitations. An 8% rate calls for evidence about who responded

- Compare respondents with the customer lists on available farm characteristics
- Report response rates separately for the offer and comparison groups
- Examine whether follow-up or weighting changes the estimate

Selective response can bias the estimated effect if it changes the comparability of the assigned groups. A larger invitation list alone does not resolve that concern

Source: AAPOR, *Response Rate Overview*

Interpreting estimates

Precision after an 8% response rate

At an 8% response rate, the study observes about **51 farms per group**. Suppose the estimated difference is **−2 acre-feet** and the standard deviation is 8 in each group. The approximate standard error is

$$SE \approx \sqrt{\frac{8^2}{51} + \frac{8^2}{51}} = 1.58$$

An approximate 95% interval is **−2 ± 1.96(1.58) = [−5.1, 1.1]** acre-feet. The data remain compatible with no effect and with reductions larger than the 2-unit effect used in the power calculation

The smaller completed sample explains both the fall to 24% power and the wide interval. The interval describes sampling uncertainty among respondents. Assess selective response using the comparisons on the previous slide

Why consider an equivalence test?

Suppose the estimated reduction is not statistically distinguishable from zero. The district is considering ending the report program

The 95% interval **[−5.1, 1.1]** still includes reductions greater than 2 acre-feet. The study has not ruled out a benefit large enough to justify the program

To support a claim that the effect is negligible, the team must specify what size of change would matter and test whether the data rule out effects that large

Equivalence testing addresses that question

Source: Kim (2019), Section 3.7

Equivalence test for the water-use report

1. Before analysis, define **−1 to 1 acre-foot** as a negligible change in monthly use
2. For a 5% equivalence test, calculate a **90% confidence interval**. The entire interval must lie inside the negligible range
3. With 51 respondents per group, the approximate interval is **[−4.6, 0.6]** acre-feet

The interval extends below −1. The data leave open reductions larger than 1 acre-foot, so they do not establish a negligible effect

Source: Kim (2019), Section 3.7

Multiple testing and false positives

The study may also test whether the report changes irrigation scheduling and crop choice. Testing **three outcomes** creates three opportunities to reject a true null hypothesis

If the three tests were independent and each used a 5% level, the chance of at least one false positive would be

$$1 - (1 - 0.05)^3 \approx 0.14$$

The outcomes may be correlated, so this value is illustrative. Identify the primary outcome before analysis, report all tested outcomes, and explain any adjustment for multiple comparisons

Source: Kim (2019), discussion of multiple testing

Check-in

Report from Tuesday’s methods workshop

Students who did not receive feedback on Tuesday can report their current methods plan

References

Falk, Armin, and James J. Heckman. 2009. “Lab Experiments Are a Major Source of Knowledge in the Social Sciences.” *Science* 326 (5952): 535–38. https://doi.org/10.1126/science.1168244.

Gibson, John. 2019. “Are You Estimating the Right Thing? An Editor Reflects.” *Applied Economic Perspectives and Policy* 41 (3): 329–50. https://doi.org/10.1093/aepp/ppz009.

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Course navigation

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